

Matt Farzaneh

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EDUCATION

Western University

B.Sc. Honors Computer Science, Minor in Software Engineering

London, ON

Expected April 2028

PROJECTS

AI Quant Council (IgnitionHacks V7) | *Python, C++17, FastAPI, Next.js, OpenAI API* | [Code](#) | [Live](#) August 2026

- Turned plain-English theses into **source-linked trade ideas**, by orchestrating a **seven-agent RAG pipeline (GPT-4o-mini)** behind a **quant validator** that rejects ideas below a **Sharpe floor** or lacking **positive benchmark excess**.
- Eliminated **look-ahead bias** across a **26,000-chunk SEC filing corpus**, verified by **229 tests** that caught a bug in the leakage detector, by enforcing **point-in-time retrieval** against a **FAISS index (text-embedding-3-small)**.
- Produced **6.0% annualized excess return over SPY at 1.23 Sharpe** on the **7 top-ranked names** of a **499-name scored universe**, by building an **event-driven backtester** charging commission and **participation-rate slippage**.
- Quantified **size-dependent slippage (0.5 to 2.5 bps from 500 to 9,000 shares)** in a standalone simulator, by writing a **C++17 limit order book** and **Nasdaq ITCH 5.0 parser (pybind11)** modeling **FIFO queue position**.

Nasdaq ITCH Order Book Engine | *C++17, Python, pybind11, Databento API* | [Code](#) | [Live](#) August 2026

- Validated a **C++/Python limit order book engine** on real **Nasdaq TotalView-ITCH (MBO)** data (**Databento API**) to **0 drift across 1,000 random checkpoints (8,000 assertions)** on a **14.47M-message day**, by cross-checking levels, order counts, and exact FIFO queue position against an independent reference implementation.
- Showed a naive fills-at-mid backtest **reversed the sign of a strategy's PnL (+\$369 vs. -\$2,679, same signal and clock)**, by building an **event-driven backtester** with queue-position-aware fills, a two-sided **2 ms / 3 ms latency model**, and real Nasdaq fees, cutting the passive fill rate to **45.7%** from an assumed 100%.
- Cut the **pybind11 Python/C++ boundary to 0.88 μ s/event**, 20x inside a 20 μ s target, and hit **7.8M events/sec** batched, by profiling the crossing itself (**85% of scalar cost**) and batching it once per replay.

WorkoutDuel | *C++17, Qt 6, SQLite, CMake* | [Code](#) April 2026

- Shipped a fully offline **two-player desktop fitness app** in **C++17** and **Qt 6**, by implementing challenge generation with equipment filtering, category-based scoring, match history, and analytics over a local **SQLite** database.
- Kept gameplay and scoring logic regression-free, by automating lifecycle, persistence, and scoring tests with **Qt Test**.

EXPERIENCE

Software Engineer Intern, Data Infrastructure

May 2026 – Present

University Pension Plan Ontario

Toronto, ON

- Saved the **Public Markets Quant** team hours of recurring manual work, by building an automated **ETL pipeline** in **Python, SQL**, and **Streamlit** that extracts from **GCP, Bloomberg, Albourne**, and the internal **DataHub**, applies **CAD FX transformations**, refreshes standardized **Excel** reports, and writes results back to **GCP**.
- Standardized reporting workflows and delivered analysis-ready datasets for quantitative research and portfolio analysis, by deploying an upgraded **Streamlit** application automating investment data transformation.
- Delivered automation matched to what the desk actually needed, by gathering requirements from **business stakeholders, portfolio managers, and engineers**.

Co-Founder / Developer

June 2025 – Present

After Image Inc

Toronto, ON

- Co-founded and shipped a **B2B SaaS platform** to production on **AWS**, by building the full-stack system in **Node.js** with **role-based access control (RBAC)** separating tenant and admin permissions.
- Automated billing, invoicing, and analytics for tenants, by building a **multi-tenant dashboard** with **Stripe integration**.
- Kept tenant-scoped dashboard and reporting endpoints responsive as data grew, by designing the **MySQL** schema and **REST APIs** with indexed access paths and query tuning.

Data Engineer & Business Analyst

May 2024 – Sept 2025

K&D Consulting

Toronto, ON

- Reduced manual Excel processing by **95%** and developer intervention by **90%+**, by building a **Python automation framework** driven by **YAML** pipeline configs.
- Cut client reporting effort by **40+ hours per week** while preserving data integrity, by managing **500+ monthly financial workbook integrations** through that framework.
- Kept delivery aligned with client requirements and deadlines, by coordinating work in **Jira** using **Agile methodologies**.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C/C++, SQL

Frameworks & Libraries: React, Node.js, Flask, FastAPI, TensorFlow, PyTorch, Pandas, NumPy

Tools & Platforms: AWS, GCP, Docker, Git, GitHub, Linux, REST APIs, MySQL, PostgreSQL, MongoDB, Streamlit, Jira